Differential Gene Expression Analysis in Mice Across Primate and Modern Diets

1. Executive Summary

This report details the transcriptomic variations in mouse liver tissue when subjected to three distinct dietary conditions: a high-fiber/low-cholesterol "Chimpanzee" diet, a high-sugar/high-fat "Human Cafeteria" diet, and a high-fat "Fast Food" diet. By utilizing Differential Gene Expression (DGE) analysis and Principal Component Analysis (PCA), this analysis identifies how specific metabolic pathways are reprogrammed in response to modern versus ancestral diets.

2. Research Objectives

The study was designed to achieve the following:

- **Comparative Analysis:** Compare gene expression of mice on experimental diets against a control group (standard Pellet diet).
- **DEG Identification:** Filter genes by fold change (>1.0) and p-value (<0.05) to isolate significant changes.
- **Pathway Mapping:** Identify biological pathways that are overexpressed (upregulated) or underexpressed (downregulated) for each condition.
- **Variance Analysis:** Use PCA to isolate genes that drive the most variation across the experimental groups.

3. Methodology and Experimental Setup

The experiment employed a DNA Microarray approach with a robust sample size to ensure statistical significance.

- **Samples:** 24 total samples (3 mice per group, across 2 batches, for 4 diet types).
- **Gene Scale:** Analysis performed on 43,014 genes.
- **Data Quality:** Initial analysis (Figure 1) showed a single peak and a positively skewed distribution. While a batch effect (Figure 2) was noted (samples clustering by processing batch), the Pellet Diet control group remained a tight, reliable cluster for baseline comparison.

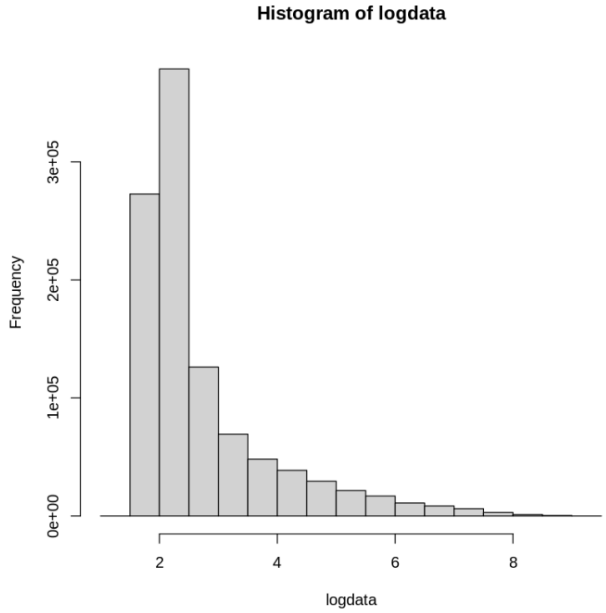

***Figure 1**: Histogram of log data*

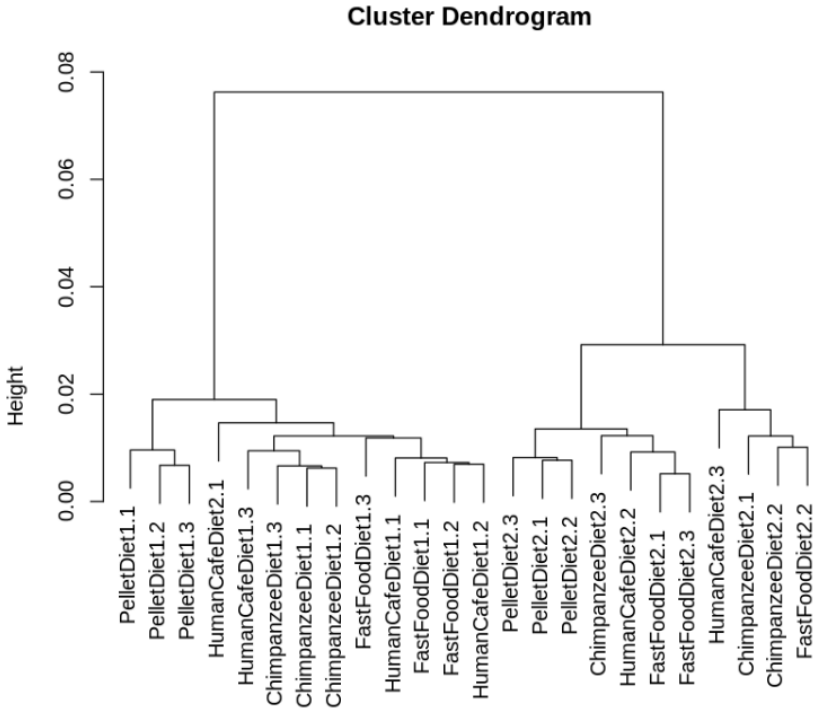

***Figure 2**: Initial clustering of data*

4. Key Findings: Differential Gene Expression (DGE)

The following table summarizes the significant genetic shifts identified across the experimental groups:

Comparison (vs. Pellet)	Total DEGs (Union)	Overexpressed	Underexpressed
Chimpanzee Diet	139	94	45
Fast Food Diet	56	26	30
Human Cafeteria Diet	74	55	19

5. Pathway Analysis

5.1 The Chimpanzee Diet (High Fiber, Low Cholesterol)

- **Upregulated:** Significant increases in cholesterol biosynthesis and lipid/sterol metabolism. This suggests a compensatory mechanism where the liver increases internal production of essential lipids absent from the diet.
- **Downregulated:** Decrease in Vitamin B3 (Nicotinamide) metabolism and long-chain fatty acid oxidation. The liver shifts away from energy-intensive fat breakdown.

5.2 The Fast Food Diet (High Fat)

- **Upregulated:** Lipid and sterol biosynthesis paths were activated, paradoxically increasing internal fat production despite high external intake. Notable activation of **Neuroimmune processes** (microglia and astrocyte activation) was observed, indicating a systemic inflammatory response.
- **Downregulated:** Inhibition of fatty acid catabolism and purine metabolism. This suggests a reduced ability to process and clear excess lipids, leading to potential liver stress.

5.3 The Human Cafeteria Diet (High Sugar and Fat)

- **Upregulated:** Acetyl-CoA/Acyl-CoA metabolism and fatty acid biosynthesis. The high sugar content appears to drive the liver to convert excess glucose into fats.

- **Downregulated:** Long-chain fatty acid metabolism. Similar to the Fast Food diet, the capacity for the liver to oxidize (burn) long-chain fats is significantly reduced.

6. Variance Analysis (PCA)

PCA (Figure 3) revealed that 83 genes in Principal Component 2 (PC2) accounted for the primary biological variation between diets. Interestingly, **66 of these genes were unique to the PCA analysis** and did not appear in the standard DEG lists. These genes are primarily involved in:

1. Regulation of Anabolic vs. Catabolic programs.
2. Acyl-CoA metabolic processes.
3. Organic acid metabolism.

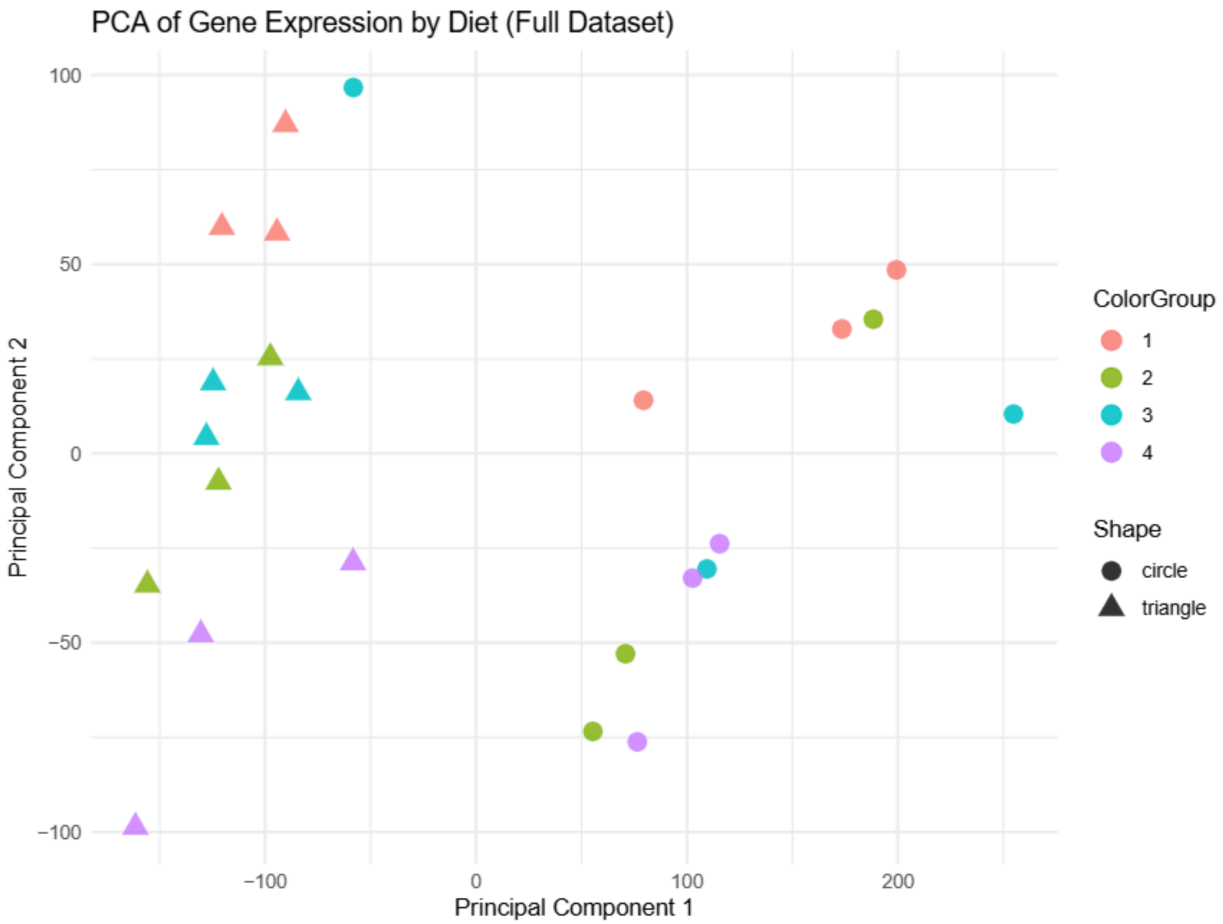

***Figure 3**: PCA results*
ColorGroup Key: 1 = Chimpanzee diet, 2 = Fast Food diet, 3 = Human Cafeteria diet, 4 = Pellet diet
Shape Key: Circle = batch 1, Triangle = batch 2

7. Conclusion

The liver displays high plasticity in response to dietary shifts. Ancestral diets (Chimpanzee) force the liver into a biosynthetic mode to create necessary lipids, while modern diets (Fast Food and Cafeteria) trigger a storage and stress mode characterized by fatty acid synthesis, reduced fat oxidation, and markers of neuro-inflammation.

Sources

1. **Primary Study:** Weng K, Hu H, Xu AG, Khaitovich P, Somel M. Mechanisms of dietary response in mice and primates: a role for EGR1 in regulating the reaction to human-specific nutritional content. PLoS One. 2012;7(8):e43915. doi: 10.1371/journal.pone.0043915. Epub 2012 Aug 24. PMID: 22937124; PMCID: PMC3427207.
2. **Dataset:** Moustafa, A. *Primates Diet Gene Expression Dataset*. Retrieved from GitHub.